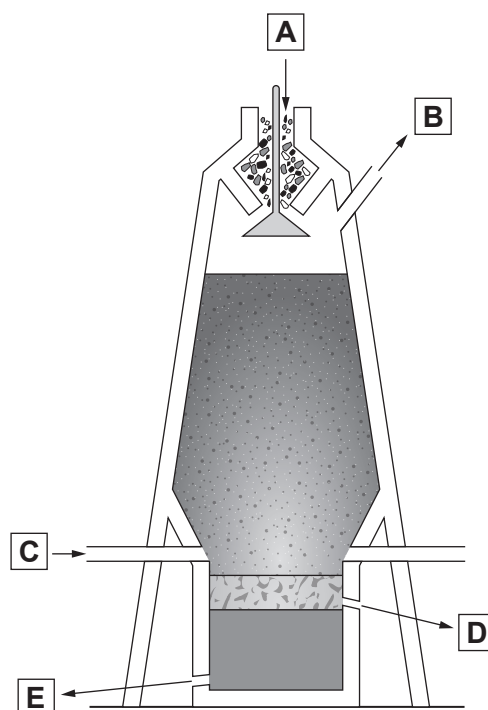


Answer **all** questions.

1. (a) The diagram shows a blast furnace which is used to extract iron.

Labels **A**, **B**, **C**, **D** and **E** show where substances enter and leave the furnace.



- (i) Give the letter, **A**, **B**, **C**, **D** or **E**, that shows where

iron ore enters the furnace .....

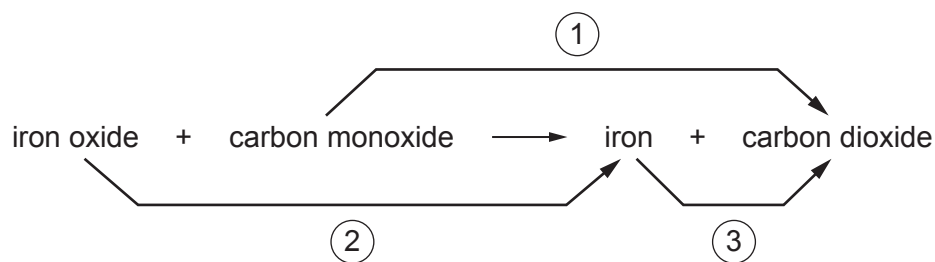
waste gases leave the furnace .....

slag is removed .....

[3]



- (ii) One of the main reactions inside the blast furnace is represented by the following equation.



Give the number of the arrow that shows oxidation.

[1]

.....

- (iii) Limestone is added to the furnace to remove impurities.

The chemical name for limestone is calcium carbonate.

Calcium carbonate contains the ions  $\text{Ca}^{2+}$  and  $\text{CO}_3^{2-}$ .

Circle the correct formula for calcium carbonate.

[1]

**CaCO****Ca<sub>2</sub>CO<sub>3</sub>****CaCO<sub>3</sub>****Ca(CO<sub>3</sub>)<sub>2</sub>**

- (iv) One of the waste gases that leaves the blast furnace is carbon dioxide,  $\text{CO}_2$ .

Put a tick (✓) in the box next to the diagram that best represents a carbon dioxide molecule.

[1]

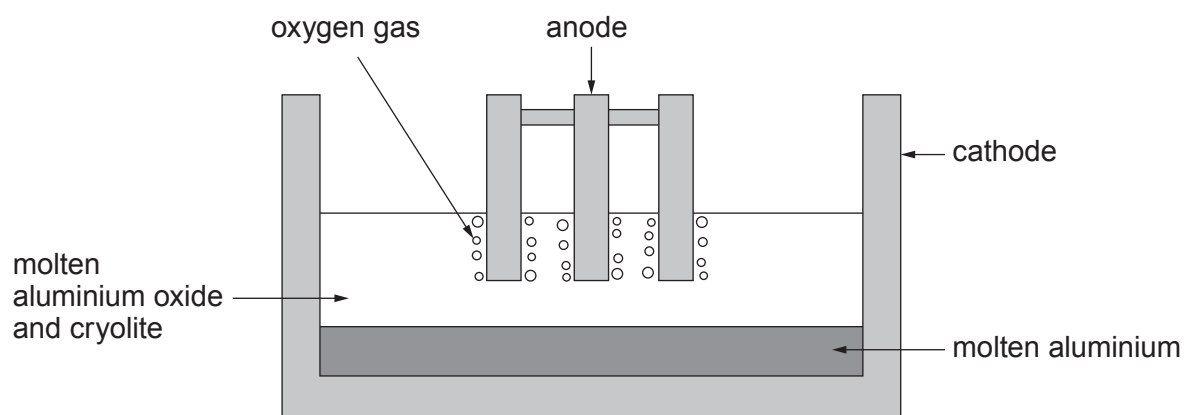

☐

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(b) The diagram shows an electrolysis cell used in the extraction of aluminium.



(i) Draw **one** line from each term to its correct definition.

[3]

| Term         | Definition                                      |
|--------------|-------------------------------------------------|
| anode        | positive electrode                              |
| electrolyte  | a substance that removes impurities             |
| electrolysis | a substance that is split up during the process |
|              | using electricity to make a compound            |
|              | using electricity to split up a compound        |
|              | negative electrode                              |



- (ii) Underline the correct word in the brackets to complete each sentence. [3]

Cryolite is added to lower the ( **density** / **melting point** / **boiling point** ) of the electrolyte.

When choosing a location for an aluminium plant in the UK, it is important to be near a port to ( **import** / **export** / **clean** ) the aluminium ore.

At the temperature inside the cell, the aluminium is produced as a ( **solid** / **liquid** / **gas** ).

- (iii) The equation for the reaction that takes place during the extraction of aluminium is given below.



Choose a number from the box to balance the equation.

[1]

|   |   |   |
|---|---|---|
| 2 | 4 | 6 |
|---|---|---|

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